



API Online

Onion Quality Assessment DoCA

AI-powered grading against AGMARK / FAQ procurement standards

Ministry of Consumer Affairs, Food & Public Distribution

Upload Onion Image

Upload a photo of a single bulb or a full sample tray. The AI will detect each bulb, measure its size, and classify defects (Rotten, Sprouted, Damaged, Undersized) against DoCA standards.

Procurement Lot & Calibration Metadata

Estimated Scale ^

Lot ID

Farmer / Vendor

Mandi / APMC Center

DOCA-8244

Ramesh Patil

Lasalgaon APMC, Nashik

Scale: ~0.635 mm/px (Uncalibrated)

Configure Calibration



1ac2f8f2-7261-4b66-abd4-6c4e0bef...

45.0 KB · image/jpeg



Ready for AI

Run AI Lot Assessment & FAQ Grading

Procurement Classes:

Sound (Grade A)

Rotten / Decay

Sprouted

Damaged (Double Split)

Undersized (<40mm)

* Double-split bulbs are classified under the official DoCA **Damaged** quota due to broken tunic and decay risk in storage.

 DoCA FAQ Standard

Inspection Certificate

Category	Count	Lot %	DoCA Limit	Status
Sound Bulbs (Healthy Grade A)	16	50%	≥ 85% Min	Fail (URS)
Rotten / Decayed	16	50%	≤ 2% Max (URS >4%)	Exceeded (URS)
Sprouted	0	0%	≤ 3% Max (URS >7%)	Within FAQ
Damaged (Double Split)	0	0%	≤ 5% Max (URS >10%)	Within FAQ
Undersized (< 40 mm)	2	6.2%	≤ 5% Max (URS >10%)	Grade II Limit

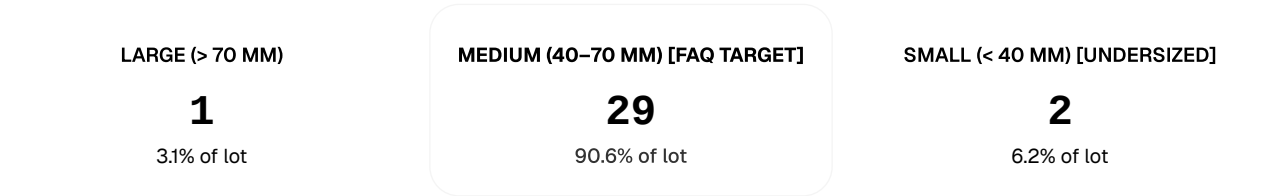
Lot Defect Composition

Healthy: 50% · Defects: 50.0%

Size Categorization

Bulb equatorial diameter distribution

Calibrated (0.635 mm/px)



Calibrated using custom scale of 0.635 mm per pixel.

Procurement Evaluation Rationale

Rule-based audit log explaining the final grade recommendation

- ✗ Rotten bulb rate of 50.0% exceeds DoCA FAQ rejection limit (4.0%). High rot risk in buffer storage.
- ✗ Sound Grade A proportion (50.0%) falls below minimum procurement threshold of 70.0%.

Anti-Double-Counting Integrity Verified

7 candidate overlapping boxes suppressed via Class-Agnostic NMS. Each physical bulb is assigned exactly one primary detection.

Generate Official Certificate

Onion Quality Assessment & Inspection Certificate

Department of Consumer Affairs · Price Stabilization Fund (PSF) Central Buffer Procurement

LOT ID:
DOCA - 8244

Date: 10 Sept 2026,
08:18 pm

HASH: 0xF915C650F82B9

FARMER / VENDOR
Ramesh Patil

PROCUREMENT MANDI
Lasalgaon APMC, Nas...

LOT SAMPLE SIZE
32 Bulbs / 50 kg

INSPECTION STANDARD
AGMARK / DoCA FAQ
2026

URS (UNDER REJECTION STANDARD) Unfit for Long-term Storage

Reject Lot - Unfit for Central Buffer Stock

Lot failed DoCA FAQ procurement standards (Defects: 50.0%). High risk of rotting and shrinkage during warehouse storage.

SOUND BULB SHARE

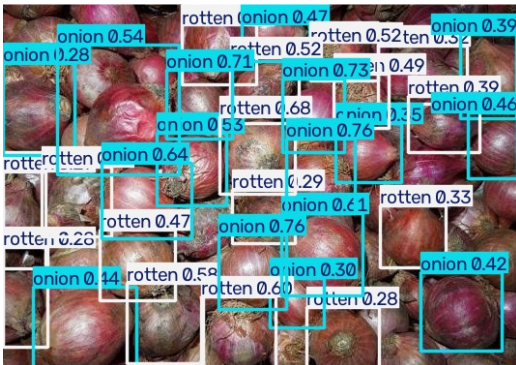
50%

DoCA Target: ≥ 85.0%

PHOTOGRAPHIC & COMPUTER VISION EVIDENCE



Raw Sample Tray Photograph



AI Annotated Detections (Class-Agnostic NMS Deduplicated)

DEFECT BREAKDOWN & FAQ PERMISSIBLE TOLERANCE ANALYSIS

Category	Detected Bulbs	Lot Share (%)	DoCA FAQ Limit	Compliance Status
Sound Bulbs (Healthy Grade A)	16	50%	≥ 85.0% Min	Fail (URS)
Rotten / Decayed	16	50%	≤ 2.0% (Max 4.0%)	Exceeded (URS)
Sprouted	0	0%	≤ 3.0% (Max 7.0%)	Within FAQ
Damaged (Double Split)	0	0%	≤ 5.0% (Max 10.0%)	Within FAQ
Undersized (< 40 mm)	2	6.2%	≤ 5.0% (Max 10.0%)	Grade II Limit

Total Lot Verified Count	32	100.0%	—	Verified (0 Duplicates)
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SIZE CATEGORIZATION (CALIBRATED (0.635 MM/PX))

LARGE (> 70 MM)	MEDIUM (40–70 MM) [FAQ TARGET]	SMALL (< 40 MM) [UNDERSIZED]
1	29	2
3.1% of lot	90.6% of lot	6.2% of lot

EVALUATION RATIONALE & DOCA PROCUREMENT RULES

- ✗ Rotten bulb rate of 50.0% exceeds DoCA FAQ rejection limit (4.0%). High rot risk in buffer storage.
- ✗ Sound Grade A proportion (50.0%) falls below minimum procurement threshold of 70.0%.

COMPUTER VISION AUDIT TRAIL	SCALE CALIBRATION STANDARD
Model: YOLOv8 Onion Precision v2 · Agnostic NMS (IoU: 0.45)	Status: Calibrated (0.635 mm/px)
Deduplication: 7 overlapping candidate boxes suppressed	Conversion Factor: 0.635 mm/px
Mean AI Confidence: 47%	Integrity: Verified: All 32 detected onions are accounted for with zero double-counting.

INSPECTING OFFICER SIGN-OFF	DIGITAL VERIFICATION STAMP
Dr. S. K. Sharma, Quality Inspector	DIGITALLY CERTIFIED
DoCA / NAFED Mandi Procurement Division	Signed at: 2026-09-10T14:48:59.316Z
ID: DOCA-APMC-INSP-4082	Hash: 0xF915C650F82B7

Problem Statement #26031 · Department of Consumer Affairs (DoCA)